Unit 14 Solutions

14.6 A sequential circuit has two inputs (X_1 and X_2) and one output (Z). The output begins as 0 and remains a constant value unless one of the following input sequences occurs:

(a) The input sequence $X_1X_2 = 01, 00$ causes the output to become 0.
(b) The input sequence $X_1X_2 = 11, 00$ causes the output to become 1.
(c) The input sequence $X_1X_2 = 10, 00$ causes the output to change value.

Derive a Moore state table.

Z
0 → 1
1 → 0

Let us look at a sample input sequence first.

X_1 : 0 0 1 0 1 0 0 0 0 1 0 1 0 1 0 1 0

X_2 : 1 0 1 1 1 0 1 1 0 0 0 0 0 1 0 1 0

Z : 0 0 0 0 0 1 1 1 0 0 1 1 0 0 1 1 1

Let us derive a Moore state graph next.

4 states:

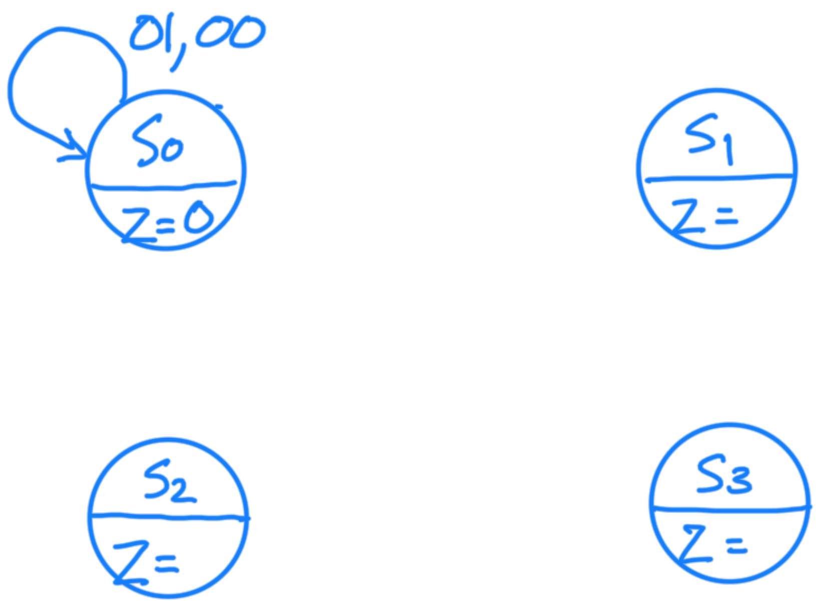
S_0 : $Z=0$

S_1 : $Z=0$, but half-way to $Z=1$

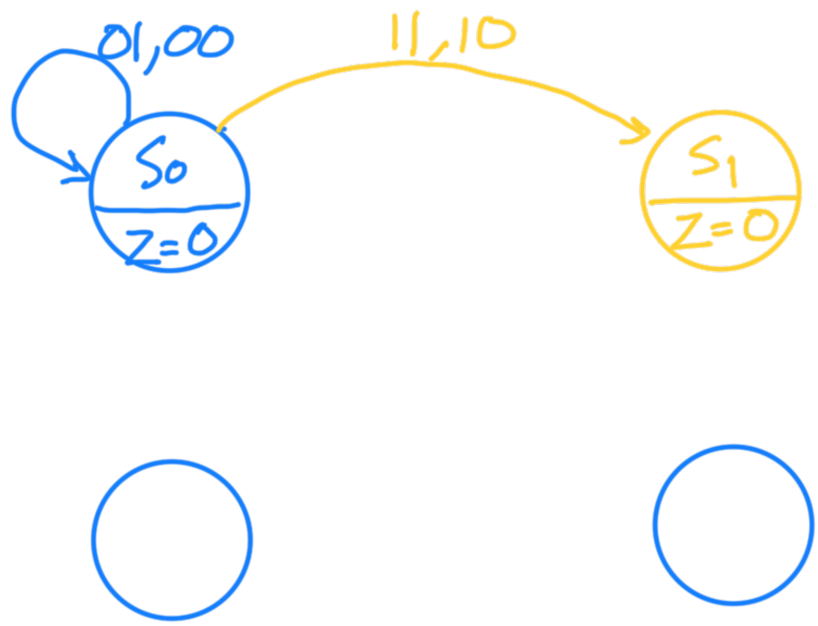
S_2 : $Z=1$

S_3 : $Z=1$, but half-way to $Z=0$

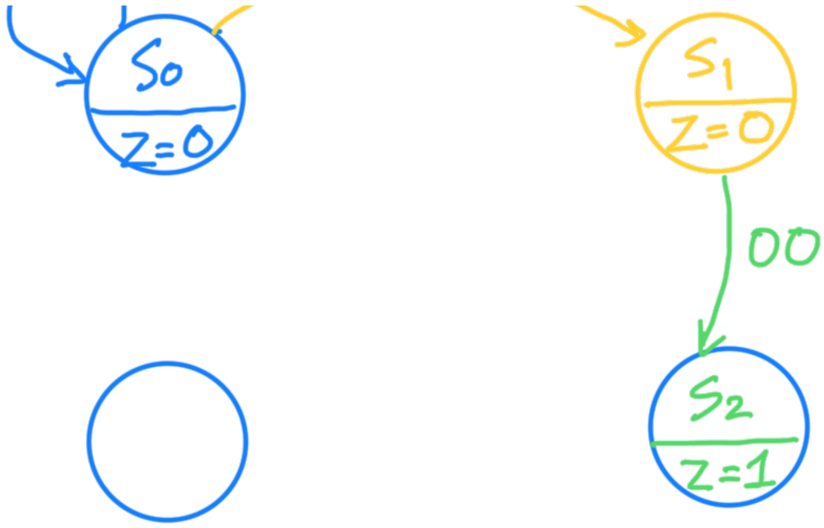
i)



ii)

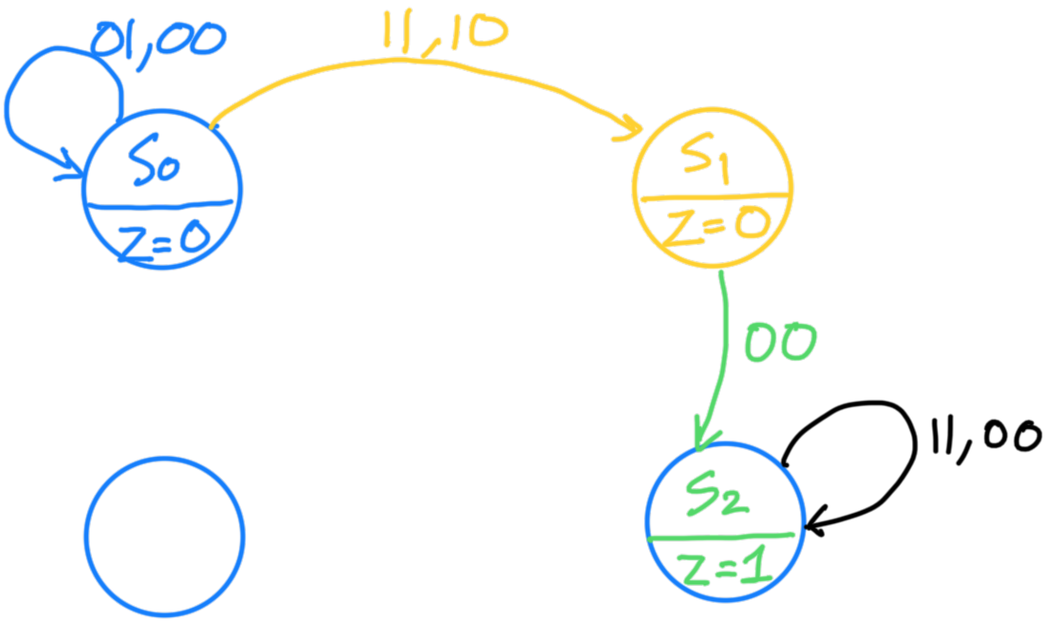


iii)



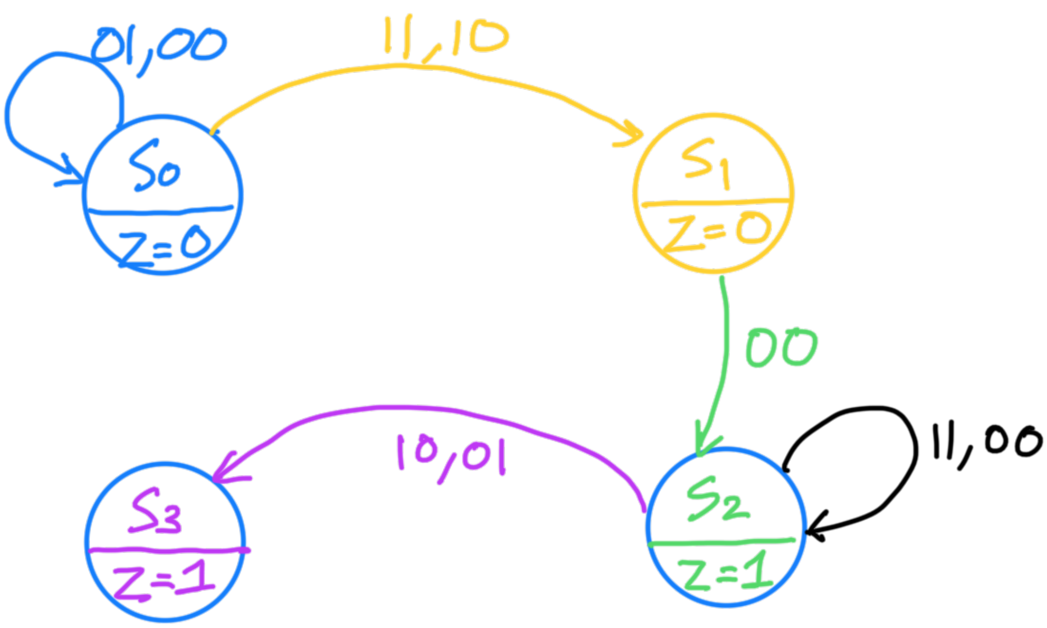
So
Z=0
S1
Z=0
00
S2
Z=1

iv)



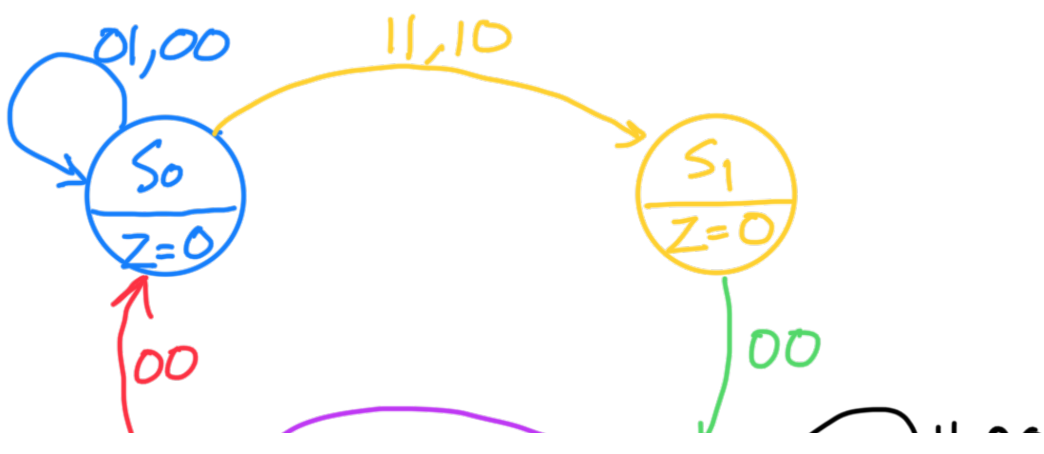
01,00
11,10
So
Z=0
S1
Z=0
00
11,00
S2
Z=1

v)

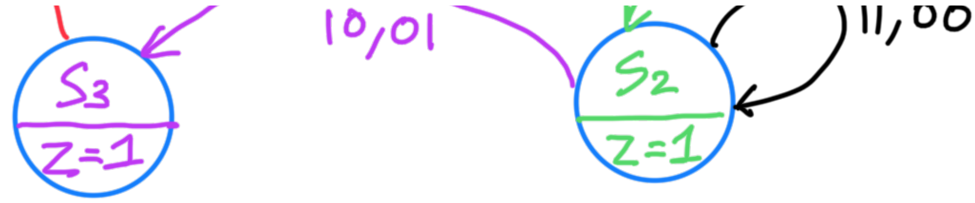


01,00
11,10
So
Z=0
S1
Z=0
00
10,01
11,00
S3
Z=1
S2
Z=1

vi)

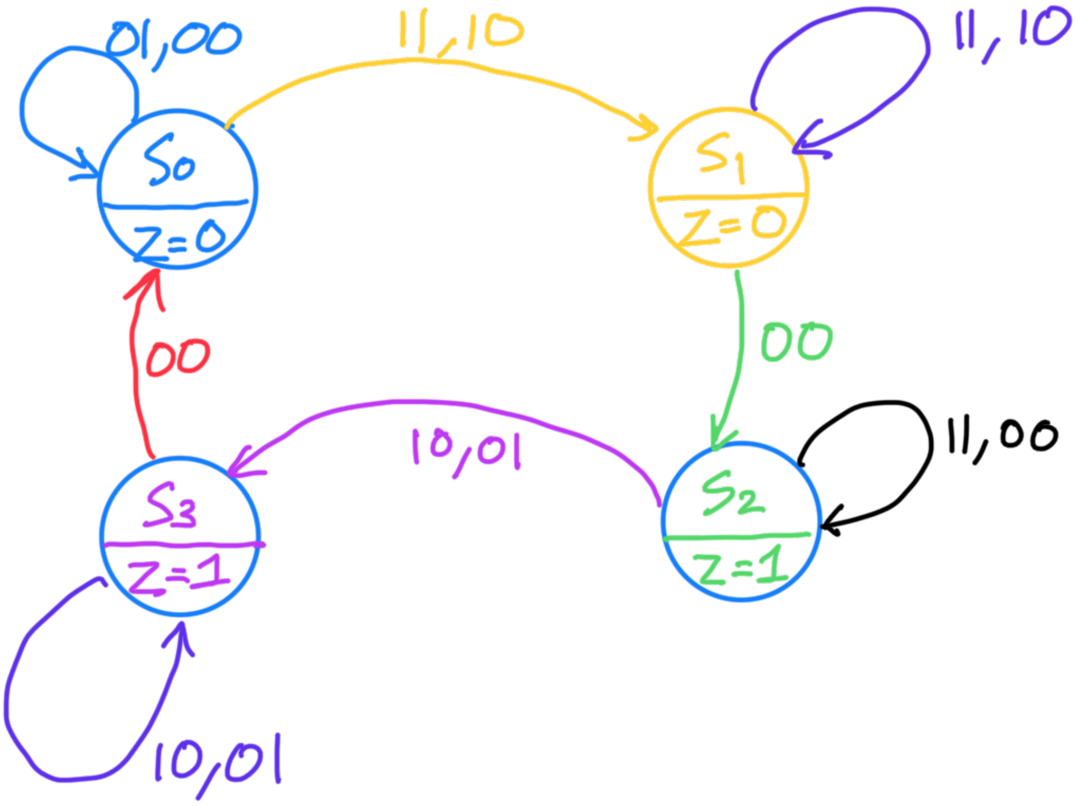


01,00
11,10
So
Z=0
S1
Z=0
00
00

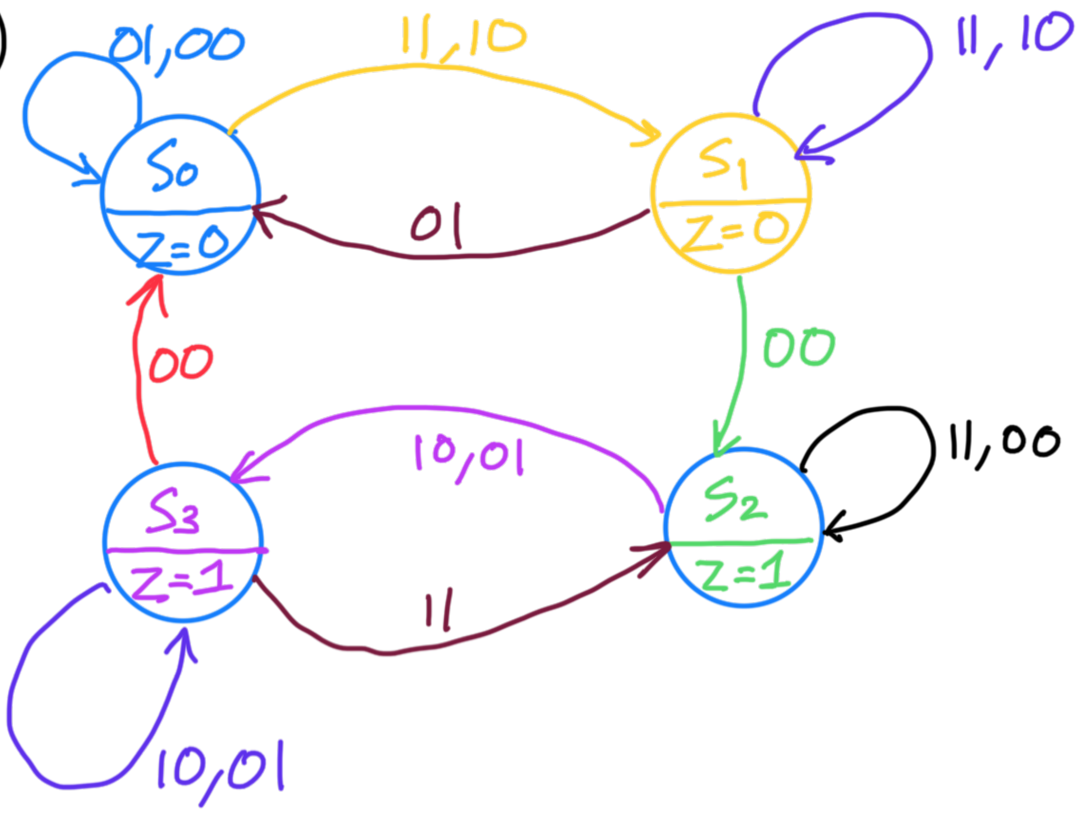


The graph isn't complete yet. There should be one arrow for each combination of X_1X_2 (00/01/10/11) for each state

vii)



viii)



X_1 : 0 0 1 0 1 0 0 0 0 1 0 1 0 1 0 1 0
X_2 : 1 0 1 1 1 0 1 1 0 0 0 0 0 1 0 1 0
Z : 0 0 0 0 0 1 1 1 0 0 1 1 0 0 1 1 1

We can now go back and see if our state graph generates the correct output Z

State table

Present state	Next state $X=00$	$X=01$	$X=10$	$X=11$	Z
S_0	S_0	S_0	S_1	S_1	0
S_1	S_2	S_0	S_1	S_1	0
S_2	S_2	S_3	S_3	S_2	1
S_3	S_0	S_3	S_3	S_2	1

14.7 A sequential circuit has one input (X) and one output (Z).
Draw a Mealy state graph for each of the following cases:

(a) The output is $Z = 1$ iff the total number of 1's received is divisible by 3. (*Note*: 0, 3, 6, 9,… are divisible by 3.)

(b) The output is $Z = 1$ iff the total number of 1's received is divisible by 3 *and* the total number of 0's received is an even number greater than zero (nine states are sufficient).

a) Let us look at a sample sequence first

X : 0001101111001010110111011
Z : 1110001001110000100010001

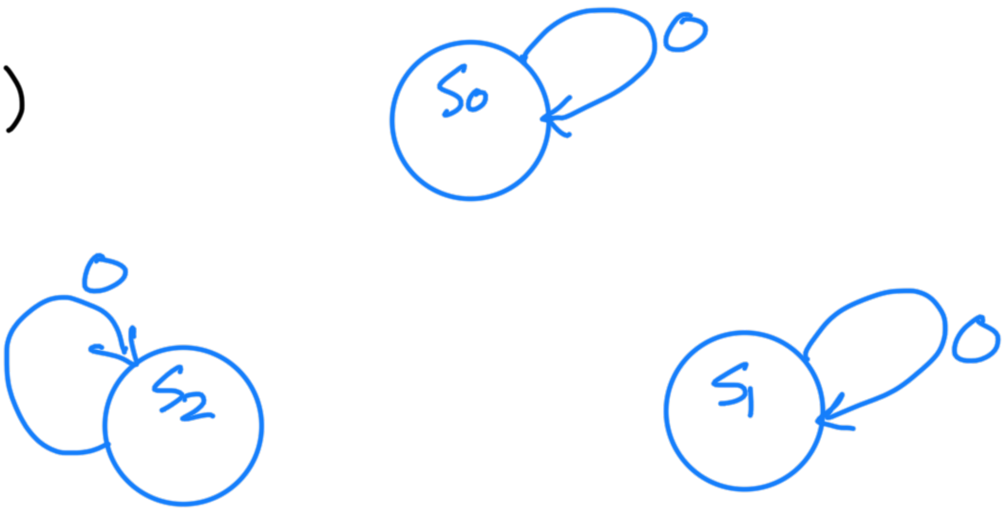
Three states

S_0 : #1s mod 3 = 0 (0, 3, 6, 9 ....)

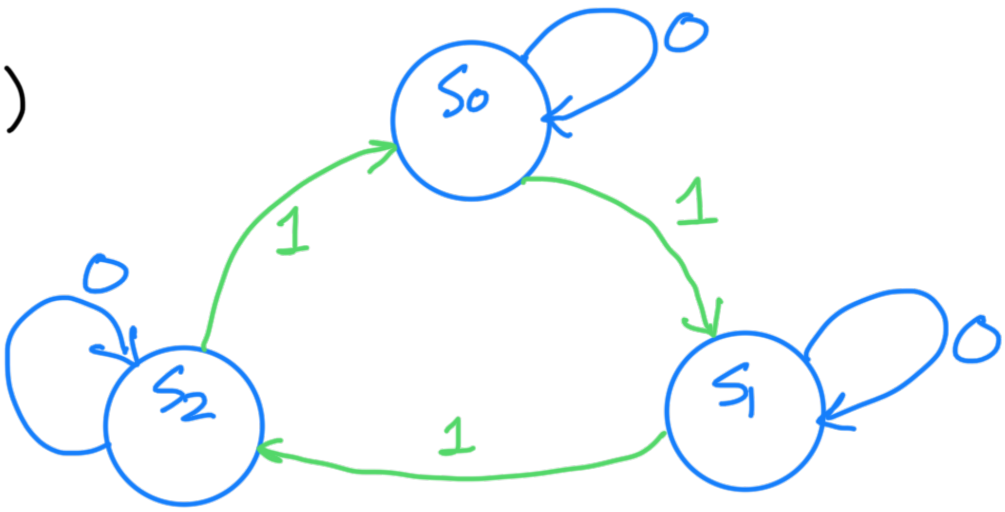
S_1 : #1s mod 3 = 1 (1, 4, 7, 10 ...)

S_2 : #1s mod 3 = 2 (2, 5, 8, 11 ...)

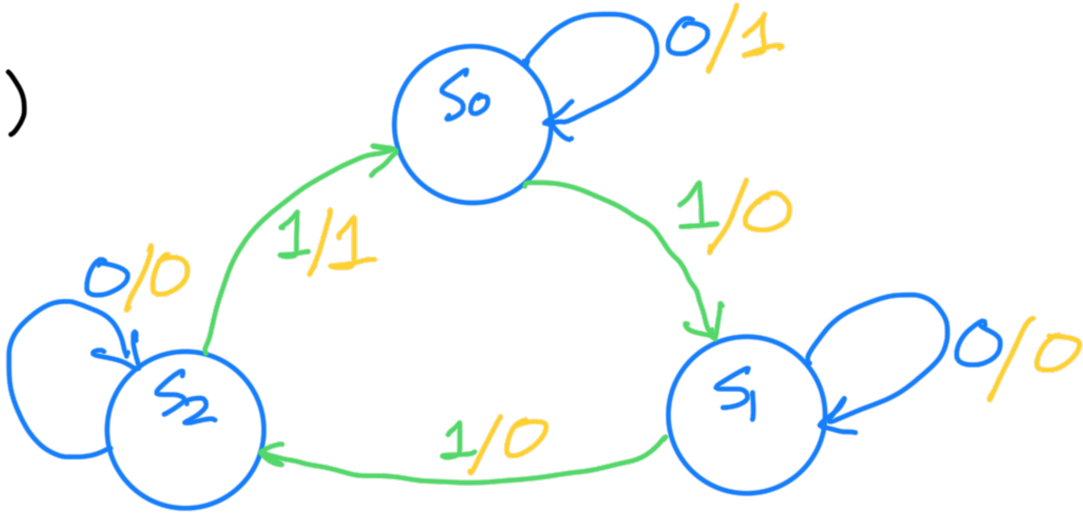
i)



ii)



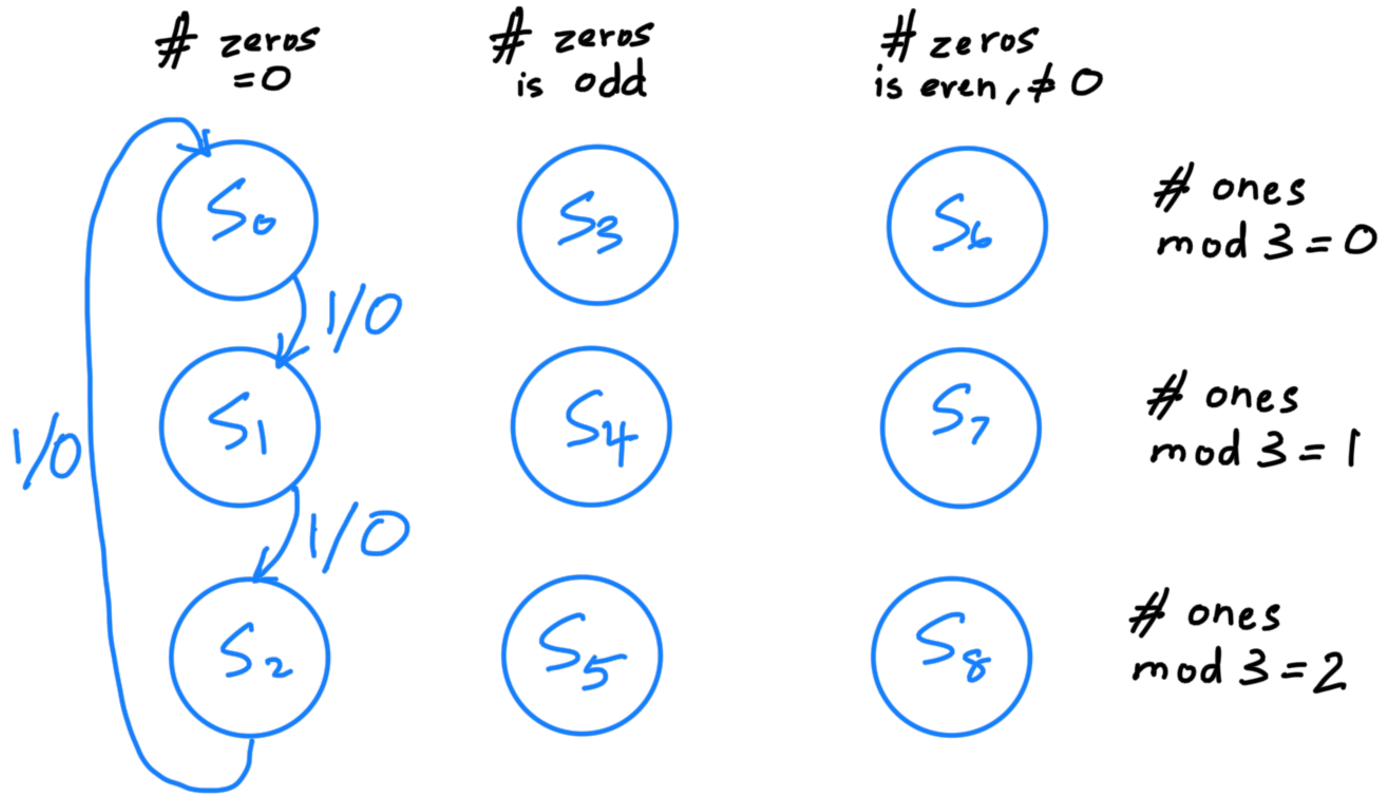
iii)



b) 9 states

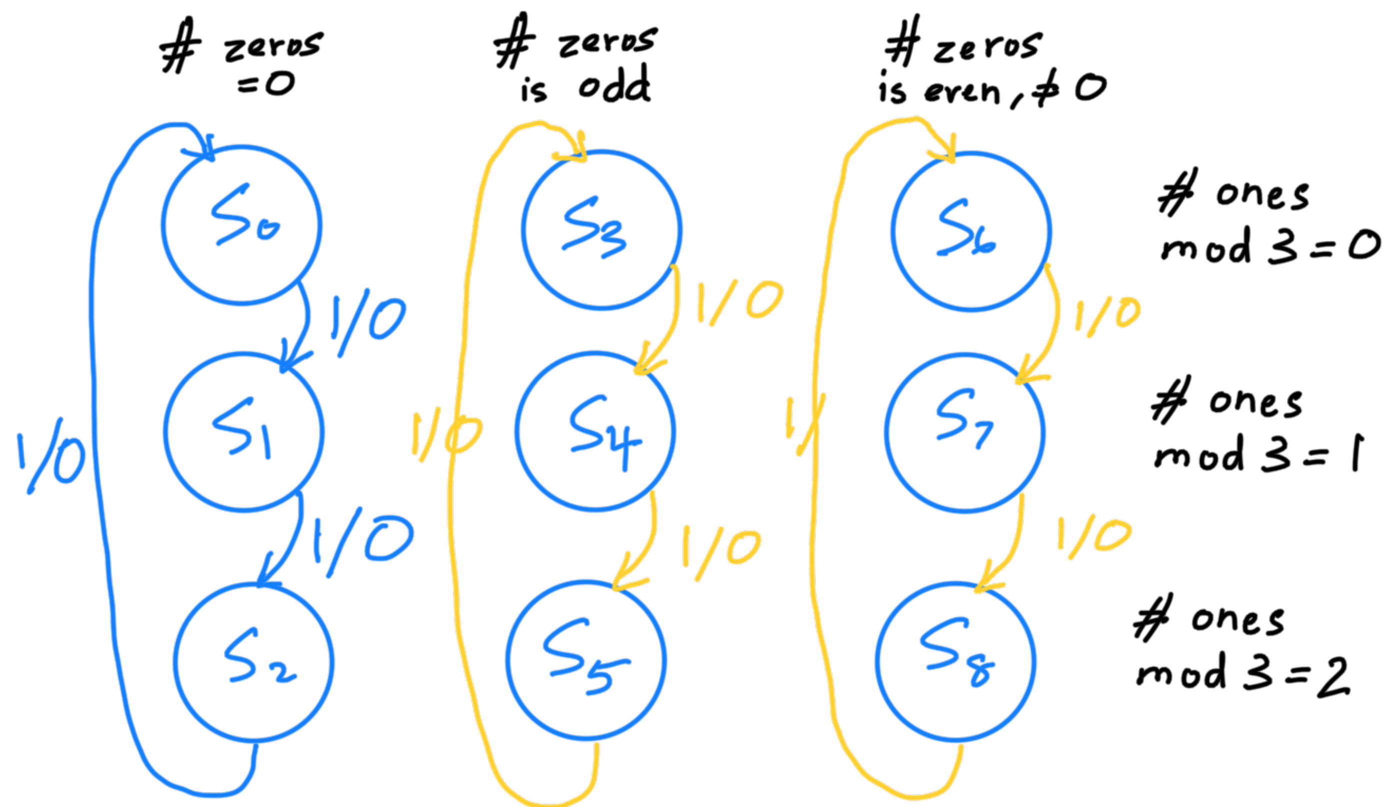
The problem says the output Z is equal to 1 if and only if the number of 1s received is divisible by and the number of 0s received is an even number greater than 0. It also says that 9 states are sufficient. Reading the problem statement, it appears that we have to distinguish between even numbers 2,4,6,8 on the one hand vs. 0 on the other. This means that there should be three sets of states. One, for when the number of 0s received is zero, two for when the number of 0s received is an odd number, and three for when the number of 0s received is an even number greater than 0. Each set of states should have 3 states each – when the number of 1s received when divided by 3 gives a remainder of 0, when the number of 1s received when divided by 3 gives a remainder of 1, and when the number of 1s received when divided by 3 gives a remainder of 2. So that’s how we have 3*3=9 states.

i)



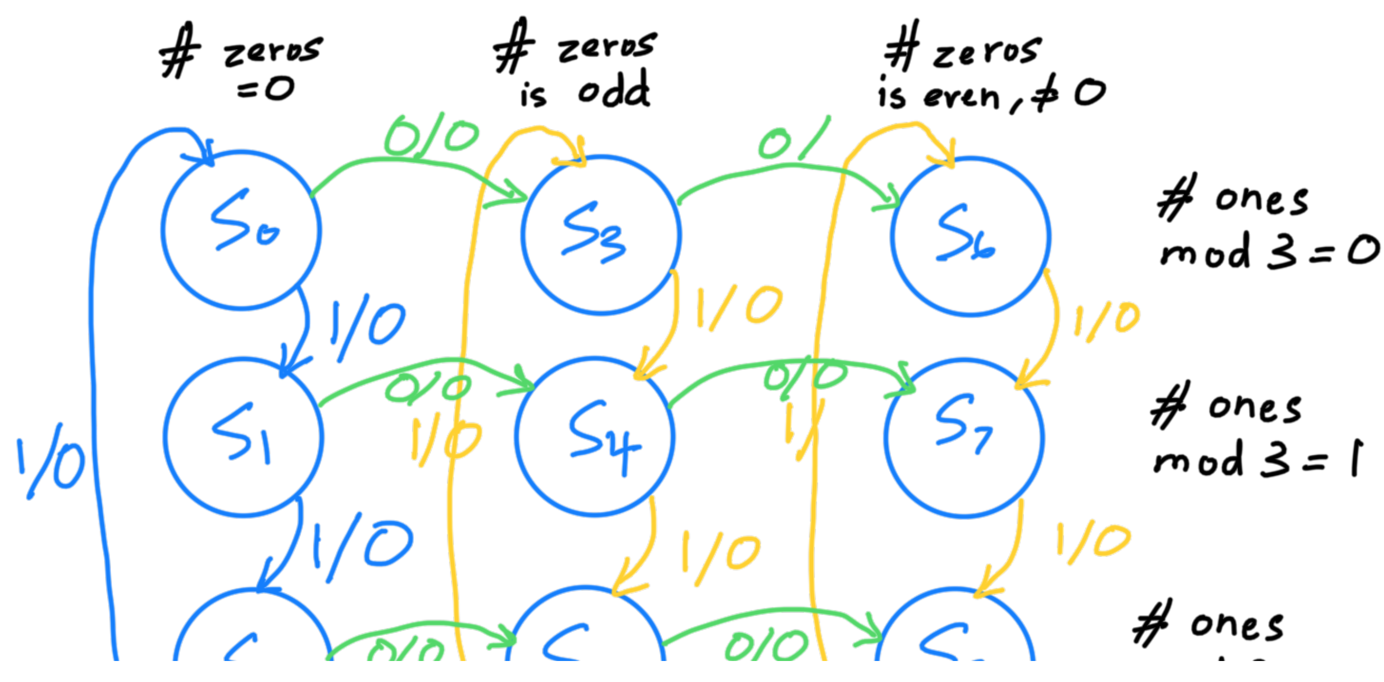
zeros =0
zeros is odd
zeros is even, ≠ 0
S0
S1
S2
S3
S4
S5
S6
S7
S8
1/0
1/0
1/0
ones mod 3 = 0
ones mod 3 = 1
ones mod 3 = 2

ii)



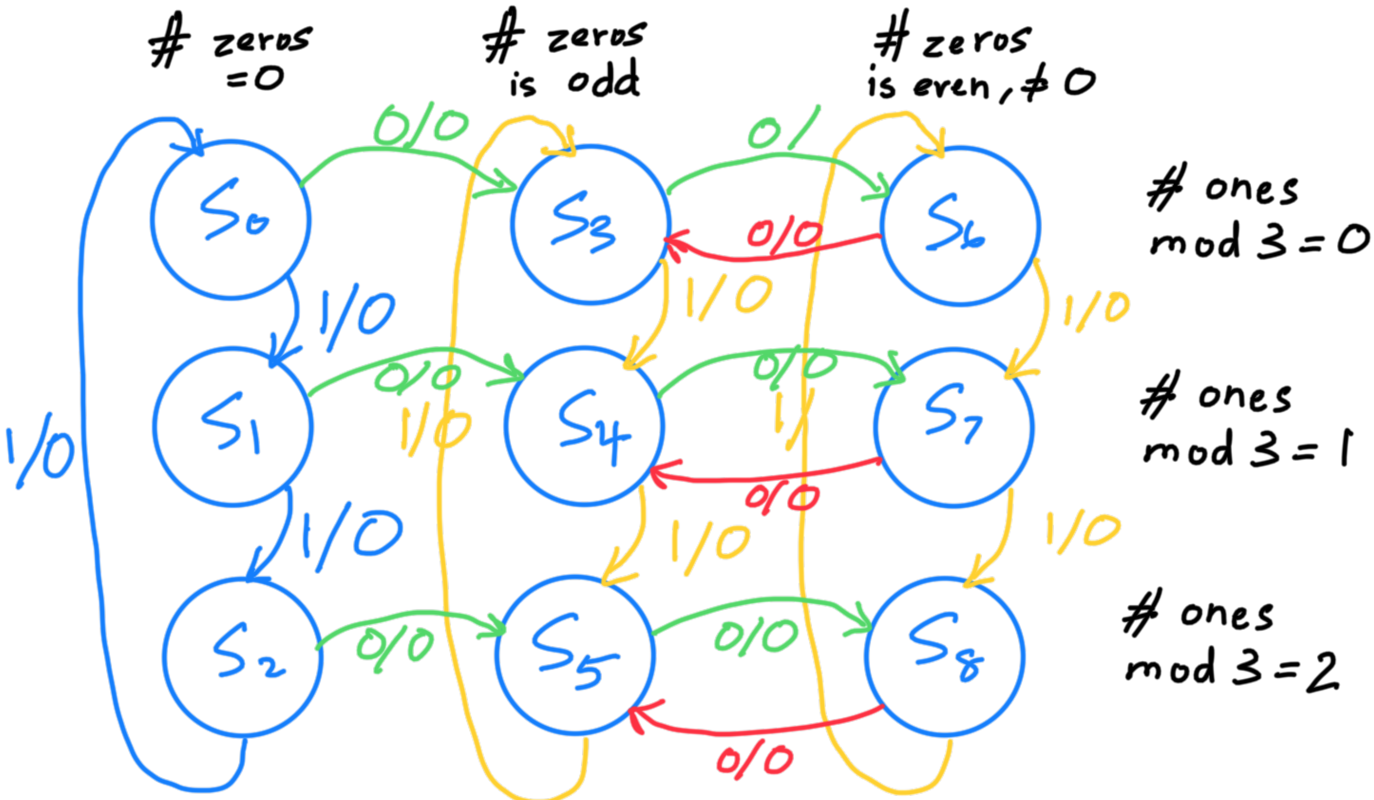
zeros =0
zeros is odd
zeros is even, ≠ 0
S0
S1
S2
S3
S4
S5
S6
S7
S8
1/0
1/0
1/0
1/0
1/0
1/0
1/0
1/0
1/
ones mod 3 = 0
ones mod 3 = 1
ones mod 3 = 2

iii)

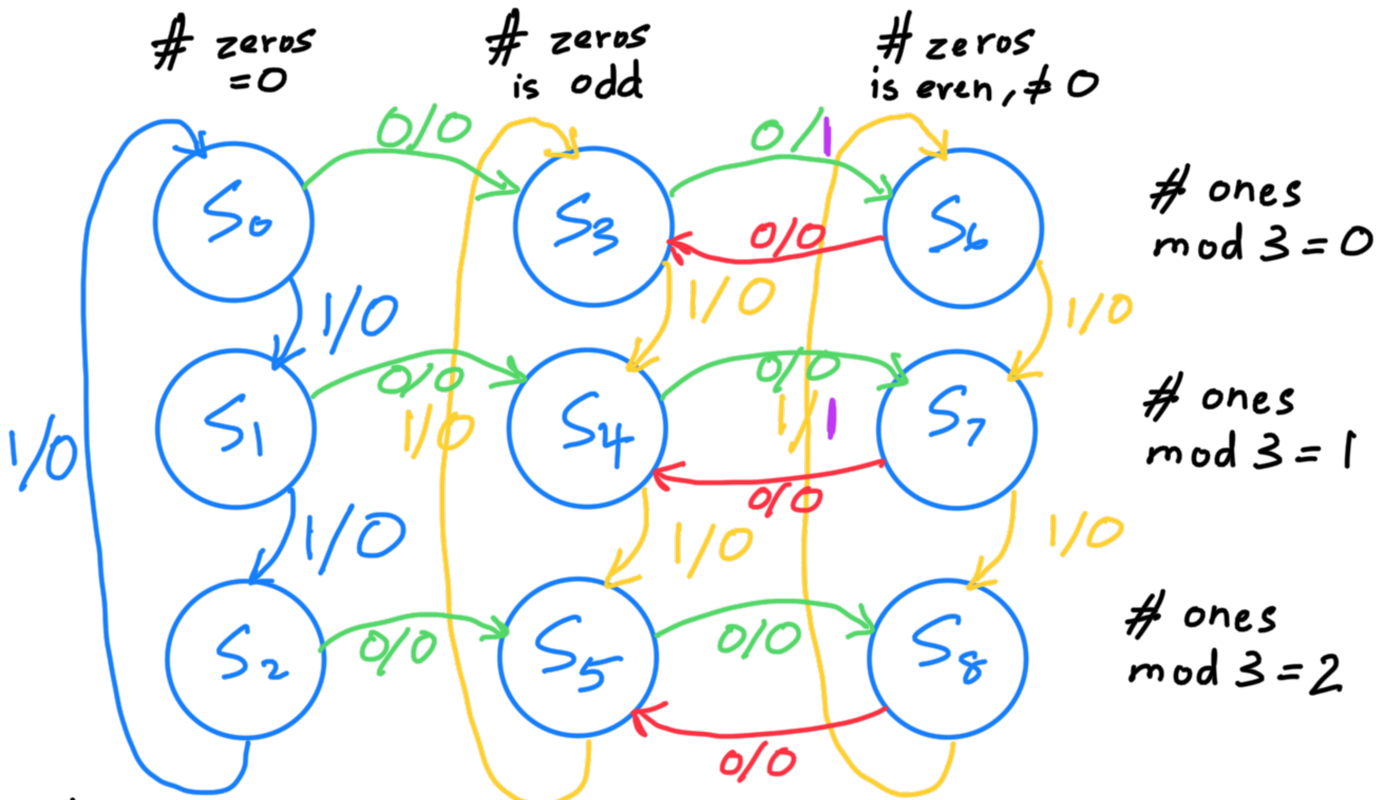


zeros =0
zeros is odd
zeros is even, ≠ 0
S0
S1
S3
S4
S6
S7
0/0
0/
1/0
1/0
1/0
0/0
0/0
1/0
1/0
1/0
1/0
1/
1/0
0/0
0/0
ones mod 3 = 0
ones mod 3 = 1
ones

iv)



v)



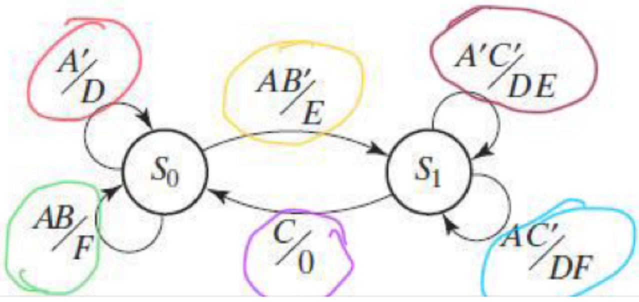
First, make sure each state is complete, i.e. there is exactly one arrow leaving each state for every possible input, i.e. 2 arrows in this problem.

Let's check with sample sequence

X: 1111011000111010

Z: 0000000101000001

14.10 For the following state graph, construct the state table, and demonstrate that it is completely specified.



For completeness, the sum of the arrows leaving each state should be =1. And, the product of each pair of arrows should be = 0.

This is a Mealy machine.

	Next state ABC= 000	001	010	011	100	101	110	111	Z ABC= 000	001	010	011	100	101	110	111
S_0	S_0	S_0	S_0	S_0	S_2	S_2	S_0	S_0	D	D	D	D	E	E	F	F
S_1	S_1	S_0	S_1	S_0	S_1	S_0	S_1	S_0	$A'C'$	0	$A'C'$	0	DF	0	DF	0

For S_0,

$$A' + \underbrace{AB' + AB}$$
$$= A' + A \quad \text{Uniting theorem}$$
$$= 1$$
$$\underbrace{A' \cdot A}B' = 0$$
$$\underbrace{A' \cdot A}B = 0$$
$$A\underbrace{B' \cdot AB} = 0$$

For S_1,

$$C + \underbrace{AC' + A'C'}$$
$$= C + C' \quad \text{Uniting theorem}$$
$$= 1$$
$$\underbrace{C \cdot AC'} = 0$$
$$\underbrace{C \cdot A'C'} = 0$$
$$\underbrace{AC' \cdot A'}C' = 0$$

∴ The state table is completely specified.